

Synthesis of CFB-Coal Fly Ash Clay Bricks and Their Characterisation

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Abstract The aim of this work was to test CFB-derived fly ash for its potential to get utilised in ceramic tiles-manufacturing by applying the sintering technique. The tested specimens were moulded using extrusion and fired at 1,050°C; a laboratory pilot-plant simulation of the industrial brick fabrication process was applied and the produced materials were afterwards tested for their microstructure and physical properties. Different types of clays were selected and characterised and various clay-ash mixtures were prepared. Plasticity after mixing with water as well as extrusion of the compact specimens and their drying behaviour were evaluated. Water absorption and mechanical strength of fired specimens were determined and evaluated as a function of the percentage FA content. Results showed that large-scale production of CFB FA-containing bricks is feasible, as their mechanical properties were not significantly harmed, while any possible detrimental effect on the other properties of the synthetic bricks appeared to be relatively restricted.

Keywords Coal fly ash · Circulated Fluidised Bed (CFB) combustion · Extrusion · Electron microscopy · Clays

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Introduction

Fluidised bed combustion is becoming a common technology for clean power production. The use of an advanced part of this technology, Circulating Fluidised Bed Combustion (CFBC) of coal for power generation, is also rapidly growing. This is mainly due to the possibility of reducing SO₂ and NO_x contents in flue gases, as well as economically burning a wide range of fuels, while meeting strict emission control regulation [1–4]. As the rate of implementation of Circulated Fluidised Bed (CFB) technology for the combustion of solid fuels gets continuously bigger, the amounts of the globally produced CFB-coal fly ash (FA) steadily increase. However, only few studies have used such FA to date for applications of commercial interest [1, 5].

An increasingly urgent environmental priority nowadays is recycling by-products and waste materials of coal-fired power stations. In fact, as such wastes occupy useful space in industrial premises, the remainder is directly discharged into fly ash ponds or landfills. Hence, both from the environmental and economic perspective, it is necessary to search for new possibilities for their utilization as industrial raw materials. Normally, less than half the amount of wastes generated in European coal-fired power stations is used as raw material for cement and concrete manufacture. Since fly ash contains alkali oxides and plant nutrients that are useful in agriculture, this material has been used as soil additive to decrease soil acidity and promote root growth and as a sulphur fertilizer. Nevertheless, a series of other applications should be investigated in order to more effectively recycle the large fly ash output. In this context, the production of conventional ceramics has the potential to be an important application of fly ash, taking into account the large quantities of raw materials needed for ceramic production [6].

The ceramic industry consumes large quantities of natural raw materials. A ceramic factory of medium size approximately uses 400 tonnes of raw materials per day. No other activity needs this high level of supply of silicates. For this reason, ceramic plants are among the most appropriate candidates for fly ash consumption. However, ceramic bodies should be designed to maximize the amount of fly ash in the initial mixture; the production would be economically unjustified otherwise. A 20–30 wt. % addition of fly ash increases the content of mullite and markedly improves the properties of stoneware products. However, earlier studies showed that the compressive strength of fly ash bricks decreases with increasing (%) fly ash content due to insufficient plasticity of bricks containing high fly ash amounts (40–60 wt. %) while water absorption increases with increasing fly ash percentage [7]. Furthermore, mixtures containing over 30 wt. % fly ash are difficult to get extruded and often some of the formed green bodies break, due to the deteriorating green strength. On the other hand, it has been reported by other studies [8] that bricks containing 50 wt. % Class F fly ash -in combination with an additive- met the commercial specifications, as they presented: (a) higher fired compressive strength and heat-insulation capacity and (b) better colour and physical appearance than bricks without fly ash. Moreover, they showed an acceptable resistance to damage, thus making them effective under severe weathering conditions. Additionally, even ceramic tiles containing more than 60 wt. % fly ash (Class F) were proved, in some cases, effective [9]. It should be mentioned that the technique reported within this paper is one of the first applied for the fabrication of bricks by utilizing CFB-derived coal FA (an ash-type that normally contains high percentages of sulphur trioxide) and thus a lower (pilot) FA participation rate was applied at this particular time. On the other hand, several recent studies [10–14] report successful incorporation of Pulverized Coal Combustion-derived FA (as well as Coal Gasification FA) in clay mixtures, in the field of conventional-sintered bricks and tiles manufacturing, leading to final products with a mechanical performance strongly dependent upon the type of ash-free body and the properties of the utilized FA.

Class C (according to ASTM C618) FA is cementitious and has hydraulic attributes due to its high calcareous content (CaO) content; CaO acts as a natural binder in the concrete mixture [15–18]. The better cementitious properties of Class C (highly calcareous) fly ash compared to class F fly ash (siliceous, CaO content <10 in weight per cent) are considered to be a disadvantage for manufacturing fired bricks or generally ceramic products [8, 18] (although they have been successfully utilised as reinforcement materials in metal-based composites [19]). In the case of Ca-rich fly ash (in the form of either CaO or Ca(OH)₂), pyretic sulphur can be captured from clays

reducing air pollution. On the other hand, calcium oxides and sulphates cause chalky deposits on the fired brick, a phenomenon termed as “scrumming” [20].

Materials and Methods

Fly Ash

The fly ash samples utilised in this work were derived from a 0.1 MW Circulating Fluidised Bed (CFB) boiler firing South African bituminous coal with the addition of limestone for sulphur capture. Boiler was operated by VTT (the Technical Research Centre of Finland) at atmospheric pressure. Inert bed material utilised in the experiments was quartz sand. Three clay samples from different deposits in Central Greece were selected and characterised. These clays are considered representative of the main types of clayey raw materials used by the ceramic industry (denoted as PAL, AML and MA2). AML is the most siliceous of three clays and hence the less calcareous; SiO₂ wt. % in PAL and MA2 is normally similar (as the result of analogous quartz content), while PAL normally contains approx. an additional 5% Al₂O₃ in weight than the latter. This tendency was also confirmed by experimentally testing clays within the framework of the current study (Tables 1, 2). The chemical composition of the raw materials was determined and presented as metal oxide composition and the amount of unburned carbon was measured as the Loss on Ignition (LOI) value. The major elements of the raw materials were determined with a Spectro X-Lab 2000 Energy Dispersive X-ray Fluorescence (XRF) spectrometer applying the samples in a pressed powder form. Mineralogical composition of raw materials and clay bricks was defined by X-Ray Diffraction (XRD) procedure using a Siemens

Table 1 Chemical composition (wt. %) of the raw materials as determined by XRF-analysis

Composition (%)	Raw materials			
	PAL	AML	MA2	CFB Fly ash
SiO ₂	46.88	52.64	48.67	46.92
Fe ₂ O ₃	5.61	5.22	3.82	2.97
Al ₂ O ₃	17.98	19.05	9.55	28.24
TiO ₂	0.72	0.68	0.43	0.93
CaO	7.53	3.51	11.82	8.30
MgO	8.15	5.91	8.39	0.61
SO ₃	0.01	0.00	0.00	3.24
P ₂ O ₅	0.14	0.18	0.98	2.49
Na ₂ O	1.75	2.45	2.03	0.66
K ₂ O	3.15	3.65	1.36	1.80
LOI	7.50	6.40	12.8	3.50

Table 2 Mineralogical composition of the raw materials

Detected phases	Raw materials			
	PAL	AML	MA2	CFB Fly ash
Quartz	XX	XX	XX	XX
Anhydrite	–	–	–	XX
Lime	–	–	–	X
Calcite	–	–	–	XX
Hematite	–	–	–	X
Mullite	–	–	–	X
Kaolinite	–	–	–	X
Illite	–	–	–	X
Albite	X	X	–	X
Gismondine	–	–	–	X
Gehlenite	–	–	–	X
Microcline	–	–	–	X
Tremolite	–	–	–	X
Augelite	–	–	–	X
Tridymite	–	–	–	X
Dolomite	X	–	X	–
Enstatite	–	–	X	–

XX intense presence, X weak presence, – absence

D-500 spectrometer with copper K α radiation. Thermogravimetric analysis (TGA) was conducted on the produced unfired and fired specimens to determine the amount of weight loss from each sample at various temperatures. The thermal analysis measurements were carried out using a Setaram TG-DTA 92-16 instrument. Samples were heated from 25 to 1,100°C in platinum holders, at a rate of 5°C/min in a static air atmosphere.

Synthesis and Characterisation of Bricks

Brick samples of 75, 40 and 17 mm dimensions (length, width, height) were manufactured by applying standard pilot-plant procedures [21, 22] and equipment: [RIETER] FWW-Lab for breaking, SR-Lab 300 for mixing and a KLVA-80 de-airing extruder. Firstly, clay samples were ground and mixed with water following standard procedures for brick fabrication in Greece, according to ELOT

771-1 (EN 771-1). Then, various clay-ash mixtures with fly ash content up to 20 wt. % were prepared and mixed with water, in order for a plastic mass to be formed; water-to-clays proportion of mixing was 22–78 in weight per cent. Plasticity of mass was afterwards evaluated and the water-amount demanded to form a plastic mass with acceptable workability was optimized using a thermobalance with IR radiation. Plastic extrusion was used for the preparation of specimens. After drying them in the air for 24 h and subsequently in an oven at 105°C for 48 h, the solid samples were sintered in a computer-controlled furnace. The first step was 500°C, which was reached after a 5 h-heating at the rate of 1.7°C/min, followed by further controlled heating up to 1,050°C at a rate of 4.5°C/min. The specimens remained at the maximum temperature for 15 min, and they then were furnace-cooled at room temperature. Chemistry and mineralogy of the so-produced fired bricks, denoted as Brick 0, Brick 10 and Brick 20, with no fly ash, containing 10 and 20 wt. % fly ash respectively, were examined by means of XRF and XRD respectively. Unfired green samples, denoted as Green Sample 10 and 20, having a fly ash content of 10 and 20 wt. % respectively, were also subjected to XRD analysis for the purpose of comparison. Bricks-surfaces were observed via optical microscopy and visual inspection to reveal the presence of possible cracking or micro-cracking. Furthermore, their morphology was examined using a JSM-6300 JEOL Scanning Electron Microscope (SEM), operated with the typical accelerating voltage of 20 kV. The microscope was coupled with an Energy Dispersive X-ray Spectrometer (EDS), with an Oxford Link ISIS system. Figures 1 and 2 show aspects of the plastic extrusion procedure, as well as the green products, as they were extruded.

Water absorption and flexural strength were determined on fired specimens and studied as a function of the fly ash content in weight percent. In order for the water absorption to be determined, the fired samples were weighed before and after immersion in water for 24 h. A three-point bending test was used to determine the modulus of rupture (M.O.R.) of the solid specimens using an automated Galdabini CTM/5 tester. The produced clay bricks were subjected to a freeze–thaw cycling test. The number of

Fig. 1 Pilot plant set-up for bricks-manufacturing





Fig. 2 Green, as extruded, products

applied freeze–thaw cycles was 25. In a single cycle, the temperature of the specimens cools from 6 to 15°C and then warms up to 6°C, within approximately 2.5–3 h for the entire cycle. All measurements were performed on 30 solid specimens of each composition and the average values were reported in the results.

Results and Discussion

Composition of Raw Materials

Chemical composition of clays, as well as this of CFB-FA, is shown in Table 1. The chemical data (Table 1) indicated that this particular FA is a siliceous, Class F ash according to ASTM C618. The relatively high percentages of sulphur are reasonable because of the limestone-assisted flue gases-desulphurization procedure that takes place during the combustion of coal within the CFB reactor. It should be mentioned that the total CaO content provides a general guide for selecting samples to avoid salt scum deposition on the surface of the bricks after firing [23]. In general, CaO values of up to 6 wt. % are manageable and lead to breaks devoid of scum deposition. However, although percentage CaO was over the aforementioned limit in the current case, it was still not high enough to put a concern over scum formation in the produced bricks; this assumption was later confirmed by experimental studying. FA composition is different compared to that of the clay samples, especially regarding the percentage presence of Fe, Al, Mg and P. Moreover, the clay samples do not contain sulphur. Loss on Ignition (LOI) value indicates the amount of volatile matter that would be released during firing. Table 1 shows that LOI in FA was much lower (up to 3.5 wt. %) than the respective of the clay samples used by the brick manufacturer. This is partially due to the fact that FA is a coal combustion by-product; FA has already underwent a high temperature heating process, where the most weakly

bonded and highly volatile organic materials had either been burned off or were converted to tightly bonded organic materials. Overall, CaO and LOI content of the ash material was acceptable for fired bricks-manufacturing.

The main mineralogical phases identified in FA are shown in Table 2. Quartz and lime are considered primary minerals, while gehlenite and anhydrite were formed throughout the combustion process. Regarding clays, they have a similar mineralogical composition compared to each other. The following mineralogical phases were identified: quartz, albite, dolomite and enstatite. Results of X-ray diffractograms of FA samples are summarised in Table 2. All ash-samples contain quartz and anhydrite and occasionally hematite, lime, calcite etc. Anhydrite is soluble and reacts with water to form gypsum or ettringite, an apparently non-desirable mineral in cement or concrete production [24]. On the contrary, the properties of mullite (low thermal expansion, excellent creep resistance up to high temperatures, low thermal conductivity) render it an ideal material for making ceramic bodies from fly ash [25].

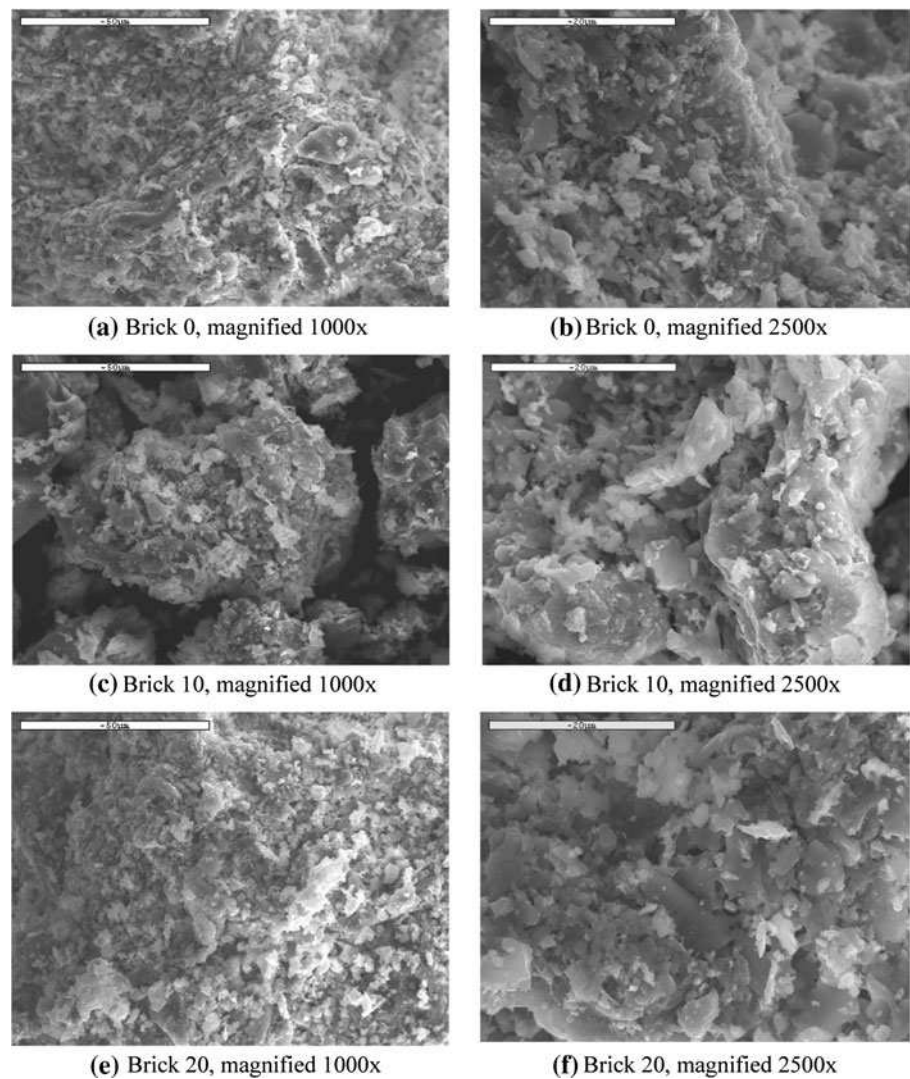
Table 3 Composition of bricks (% wt.) determined by XRF analysis

Composition	Brick 0	Brick 10	Brick 20
SiO ₂	38.62	37.95	36.83
Fe ₂ O ₃	19.71	21.12	21.2
CaO	19.95	18.19	18.16
Al ₂ O ₃	10.98	11.27	11.16
K ₂ O	4.62	4.54	4.64
SO ₃	1.75	2.24	2.52
TiO ₂	1.51	1.75	1.98
MgO	1.47	1.48	1.22
BaO	0.60	0.56	0.51
MnO	0.34	0.37	0.35
Cr ₂ O ₃	0.13	0.12	0.18

Table 4 Mineralogical composition of the examined samples

Mineral	Brick 0	Brick 10	Brick 20	Green sample 10/20
Quartz	XX	XX	XX	XX
Anhydrite	–	X	X	–
Mullite	XX	XX	XX	–
Albite	X	X	X	X
Microcline	X	X	X	–
Anorthite	X	X	X	X
Diopside	X	X	X	–
Muscovite	–	–	–	X
Clinoclere	–	–	–	X
Gypsum	–	–	–	X

XX intense presence, X weak presence, – absence

Fig. 3 Microstructure of manufactured bricks

Characterisation of Products

Although differences in plasticity were observed when fly ash was added in various amounts in the clay mixture, they did not cause severe problems in the specimen preparation. Actually, extrusion and drying behaviour of unfired bricks can be considered satisfactory, as the produced samples obtained the green strength required to ensure safe handling in the subsequent fabrication steps. Therefore, no additive was demanded to facilitate fabrication of unfired samples with or without FA addition to the clay mixture. The fired brick specimens of each composition presented the same colouring. Because of their iron oxides content, the pure clay bricks had a red-brown colour. An increasing addition of fly ash in the mixture of raw materials leads to a gradual shift to more pale colouring [25]. Phase composition of fired clay bricks is not significantly modified when incorporating FA up to 20 wt. %. In fact, the same basic

mineralogical phases, namely quartz, mullite, albite, diopside, anorthite and microcline were identified in the samples prepared both with and without FA addition. In the brick samples with FA addition, anhydrite was additionally detected, probably due to its initial presence in FA. The CaO content (Table 3), slightly decreases in fired bricks when the FA-percentage presence is increased. No free lime, which could adversely affect mechanical behaviour and even cause cracks and fissures, was detected in bricks of all the examined compositions. The mineralogical composition of the end products before and after firing is presented in Table 4. All specimens contained quartz, mullite, albite, microcline, anorthite, diopside and occasionally anhydrite.

Thermogravimetric analysis was conducted on the clay bricks and the respective green samples in order for the amount of weight loss from each sample at various temperatures to get determined. Results indicated that the

weight loss occurred for the clay bricks in the temperature range of about 51–230°C and 931–1,065°C. For the green samples the temperature range was 112–136°C, 514–671°C and 680–743°C. In general, DTA (Differential Thermal Analysis) curves enable detection of exo- and endothermic peaks (effects due to gain/loss of enthalpy) occurring in the sample when undergoing controlled heating and compared to an inert reference material [26]. In clay minerals, the endothermic peak around 100°C is due to moisture water, whereas those appearing at about 200–250°C are attributed to ‘bound’ water, or to ‘hydrated’ interlayer cations (as in swelling clay minerals). As far as the green sample is concerned, gypsum shows endothermic effects within the range of 120–160°C. The most common clay minerals such as kaolinite, illite contained herein fly ash used, are recognizable by their relatively strong endothermic effects within the range of 550–650°C (green sample). Carbonates show distinctive endothermic peaks at 840°C (calcite) and doublets at 780°C and 860°C (dolomite) in static air atmosphere, whose position may vary depending on grain size, crystal size, atmosphere and other concomitant factors [27].

The microstructural examination (Fig. 3) confirmed that the specimens with FA addition presented a higher porosity than the non-FA specimens. The presence of open pores within the form of sponge-like pores is observed at several magnifications (1,000 and 2500×). The sponge-like pores were probably formed by CaCO_3 decomposition [23]. Water absorption rates have been used as an indication of porosity of the produced specimens. The results for water absorption of fired bricks (Fig. 4) indicate an almost linear relationship between FA content and water absorption, for FA addition up to the level of 20 wt. %. This is apparently due to the formation of open pores as the percentages of subsistent increase, which was confirmed by microstructural examination as well as by visual observation of bricks. Further FA addition in the mixture surprisingly led to a slight reduction and stabilization of the water absorption rate (above 15%), which can be attributed to the possible presence of closed porosity.

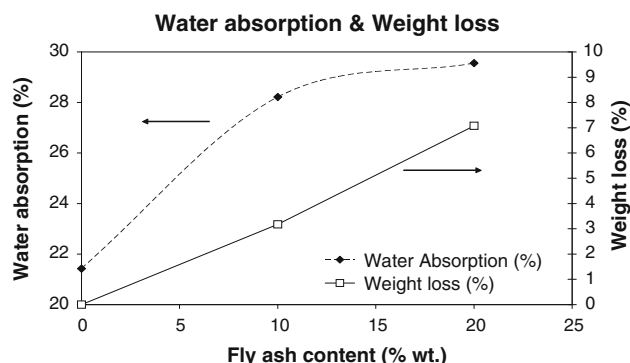


Fig. 4 Water absorption (%) and weight loss (%) of the manufactured bricks, as a function of fly ash content (%)

In the case higher relative densities were demanded, higher sintering-temperatures would have been considered. From the financial point of view, cost-reduction is achieved by producing objects of a reduced relative density. Besides, a porous microstructure offers advantages for specific applications, i.e. regarding insulation behaviour or even thermal shock-resistance, due to the improved expansion tolerance and a certain decrease in the modulus of elasticity [28]. The shrinkage, which both pure and FA-containing samples underwent during heating, appeared to be relatively restricted and remained within tolerable limits for standard brick-fabrication.

The modulus of rupture of bricks, as it was derived by the 3-point bending test, is shown in Fig. 5; Table 5 shows standard deviation of the analysis. It can be seen that flexural modulus of rupture decreases with the increase of fly ash content, both before and after the cyclic freeze–thaw test. It should be mentioned that such decrease is less pronounced for FA additions of 10–20 in weight per cent, due to the sealed porosity. Hence, mechanical behaviour drawbacks appear tolerable only for low ash additions. Actually, strength of porous ceramics strongly depends on their microstructure. Regarding porosity increase, as it reduces the load-bearing area during mechanical testing, fractures subsequently occur at lower applied loads, which results in flexural strength decrease [29].

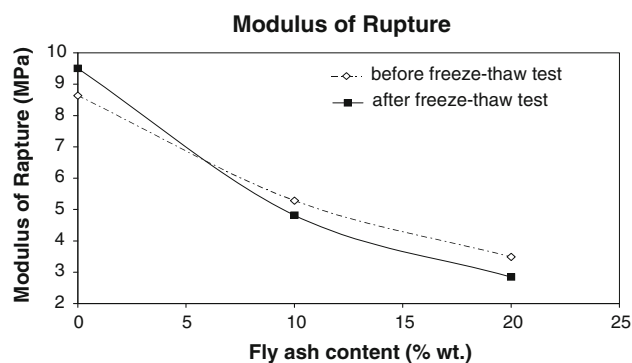


Fig. 5 Modulus of rupture of the manufactured bricks, as a function of fly ash content (%)

Table 5 Average applied load and modulus of rupture of tested specimens (incl. standard deviation)

FA wt. %	Load (kN)	Length (m)	Width (m)	MOR ^a (MPa)
Non-FA samples	1.86 ± 0.20	0.0415	0.01673	8.63 ± 0.94
10	1.27 ± 0.16	0.0423	0.0175	5.29 ± 0.67
20	0.82 ± 0.22	0.0417	0.0174	3.49 ± 0.61

^a MOR modulus of rupture

Conclusions

The main conclusions of the study can be summarized as follows:

- Clay bricks containing 10 and 20 wt. % CFB-derived coal fly ash were successfully produced using pilot-plant procedures.
- According to the results of their physico-chemical evaluation, any possible detrimental effects on the physical properties of the bricks, by using CFB FA in the mixture of raw materials, generally appeared to be relatively restricted and tolerable only for low percentages of FA incorporation.
- Low percentages of FA substitution in clay bricks is possible without significantly deteriorating the mechanical performance, while porosity progressively increases, which could be of importance in terms of heat insulating behaviour.
- For high FA contents, lower values of strength were obtained. Compared to the standards applied in the industry, i.e. for clay roofing tiles, the tiles produced are acceptable in terms of flexural strength but not of the other physical characteristics (water absorption and frost resistance), suggesting that only small amounts of fly ash (<10% wt.) could be utilized for the production of clay roofing tiles.

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